

Clustering concepts

Clustering of Data

What is clustering?

Unsupervised

Identifying Patterns

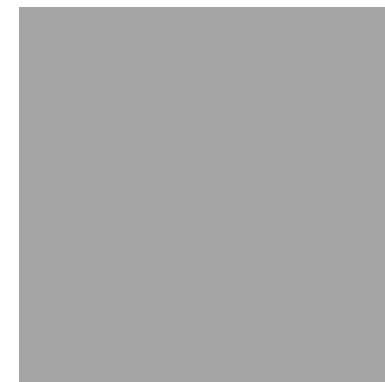
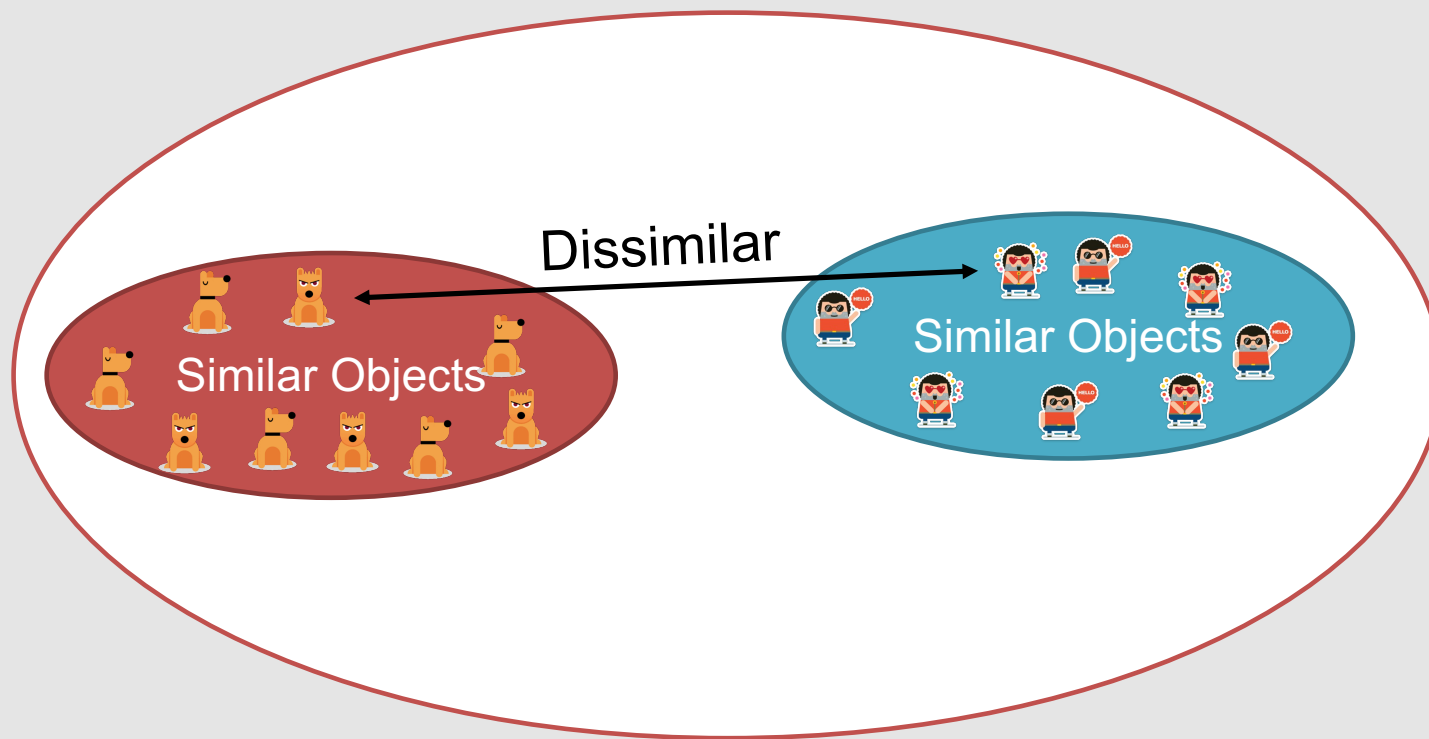
Grouping/Segmentation

Exploratory Analysis

Let's cluster this set of images



Similar/Dissimilar



Distance Metrics

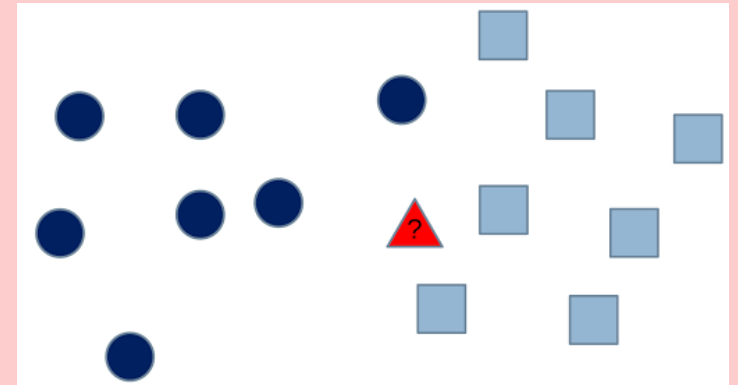
- Measuring similarity or distances between different data points is fundamental to many machine learning algorithms
 - unsupervised learning problems (i.e. K-means method in clustering)
 - supervised learning methods (i.e. K-Nearest-Neighbor)
- Distance measures are functions that define a distance $d(x_i, x_j)$ between any two data instances x_i and x_j for measuring how similar the instances are.

Distance Metrics...

- ❖ Distance measures satisfy the following three properties:
 - ❖ For any instance x_i , **distance with itself is zero**, $d(x_i, x_i) = 0$
 - ❖ For an instance pairs x_i and x_j , **the distance is non-negative and symmetric**, $d(x_i, x_j) \geq 0$ and $d(x_i, x_j) = d(x_j, x_i)$
 - ❖ Distance measure follows **triangular inequality**
$$d(x_i, x_k) \leq d(x_i, x_j) + d(x_j, x_k)$$
- ❖ Distance measures satisfying above properties are also known as Distance Metrics

Distance Metrics...

- Example 1: Nearest Neighbour Classification
 - Using distance to find the label of the new data point (the red triangle - Square or circle?)





Distance Metrics...

- Example 2: Image retrieval
 - Animal types in NUS Wide Animal dataset. Given a new image like the image of a cat, can we fetch all cat images from the dataset? (yes! with the help of distance measurements)

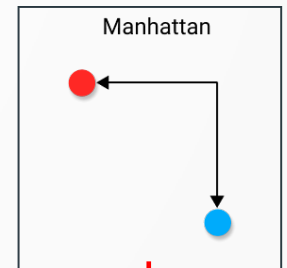
Distance Measurement Types

- Cityblock/Manhattan distance

- For any two data instances, represented by d-dimensional feature vectors x_i, x_j , their Cityblock distance is computed as:

$$d_{Cityblock}(x_i, x_j) = |x_{i,1} - x_{j,1}| + \dots + |x_{i,D} - x_{j,D}|$$

- In most cases, this results similar to the Euclidean distance
- However, the effect of **a large difference in a single dimension is dampened** (since the distances are not squared)



$$d(x_1, x_2)_{CB} = |1 - 0| + |1 - 2| + |2 - 2| + |1 - 0| + |0 - 2| = 5$$

$$x_1 = \begin{bmatrix} 1 \\ 1 \\ 2 \\ 1 \\ 0 \end{bmatrix}^T \quad \text{and} \quad x_2 = \begin{bmatrix} 0 \\ 2 \\ 2 \\ 0 \\ 2 \end{bmatrix}^T$$

Distance Measurement Types

- Euclidean distance

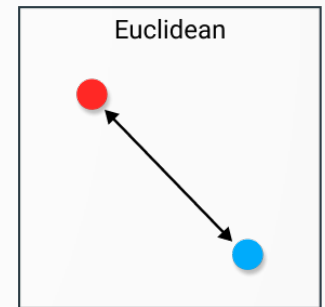
- Straight-line distance between two points in Euclidean space
- For any two data instances, represented by **d-dimensional feature vectors** x_i , x_j their Euclidean distance is

$$d_{Euclidean}(x_i, x_j) = \left((x_{i,1} - x_{j,1})^2 + \dots + (x_{i,D} - x_{j,D})^2 \right)^{1/2}$$

- For example, consider these two vectors:

$$x_1 = \begin{bmatrix} 1 \\ 1 \\ 2 \\ 1 \\ 0 \end{bmatrix} \text{ and } x_2 = \begin{bmatrix} 0 \\ 2 \\ 2 \\ 0 \\ 2 \end{bmatrix}$$

$$\begin{aligned} d(x_1, x_2) &= \left((1 - 0)^2 + (1 - 2)^2 + (2 - 2)^2 + (1 - 0)^2 + (0 - 2)^2 \right)^{1/2} \\ &= \sqrt{(1 + 1 + 0 + 1 + 4)} = \sqrt{7} = 2.65 \text{ (approx)} \end{aligned}$$



Distance Measurement Types

- Chebyshev distance

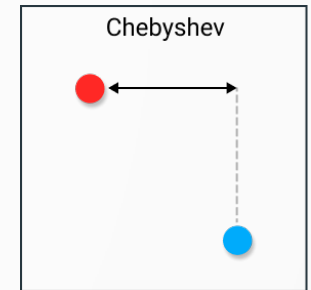
- Greatest of distances along any coordinate dimension
- For any two data instances, represented by **d-dimensional feature vectors** x_i , x_j their Euclidean distance is

$$d_{\text{chebyshev}}(x_i, x_j) = \max(|x_{i,1} - x_{j,1}|, |x_{i,2} - x_{j,2}|, \dots, |x_{i,D} - x_{j,D}|)$$

- For example, consider these two vectors:

$$d(x_1, x_2) = \max(|1 - 0|, |1 - 2|, |2 - 2|, |1 - 0|, |0 - 2|)$$

$$= \max(1, 1, 0, 1, 2) = 2$$



$$x_1 = \begin{bmatrix} 1 \\ 1 \\ 2 \\ 1 \\ 0 \end{bmatrix} \text{ and } x_2 = \begin{bmatrix} 0 \\ 2 \\ 2 \\ 0 \\ 2 \end{bmatrix}$$

Distance Measurement Types

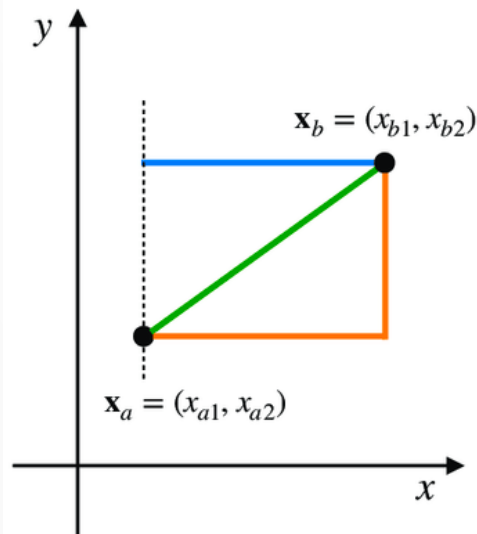
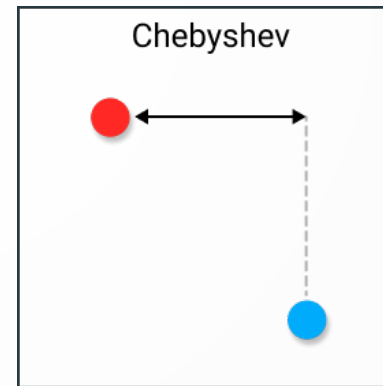
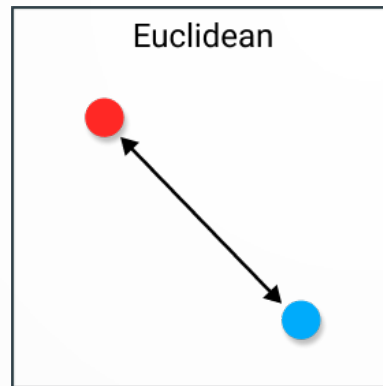
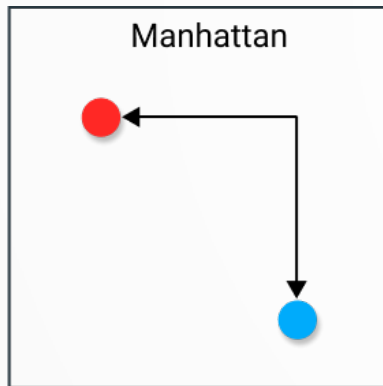
- Minkowski distance

- Defines a distance between two points in a normed vector space.
- Euclidean and Cityblock distances are 2nd and 1st normed distance between x_i and x_j .
- Minkowski distance is a **generalization** of these distances defined for any p-norm

$$d(x, y) = \left(\sum_{i=0}^{n-1} |x_i - y_i|^p \right)^{\frac{1}{p}}$$

- when $p = 1$, d is Manhattan distance
- when $p = 2$, d is Euclidean distance
- when $p = \infty$, d is Chebyshev distance

Schematic overview of popular distances



$p = 2$ Euclidean distance

$$\|\mathbf{x}_a - \mathbf{x}_b\|_2 = (|x_{a1} - x_{b1}|^2 + |x_{a2} - x_{b2}|^2)^{\frac{1}{2}}$$

$p = 1$ Manhattan distance

$$\|\mathbf{x}_a - \mathbf{x}_b\|_M = |x_{a1} - x_{b1}| + |x_{a2} - x_{b2}|$$

$p = \infty$ Chebyshev distance

$$\|\mathbf{x}_a - \mathbf{x}_b\|_\infty = \max\{|x_{a1} - x_{b1}|, |x_{a2} - x_{b2}|\}$$

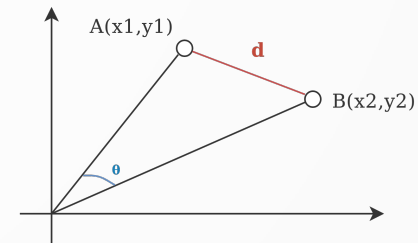
Distance Measurement Types

- Cosine distance

- For any two data instances, represented by d-dimensional feature vectors

x_i, x_j

$$d_{\text{Cosine}}(x_i, x_j) = 1 - \frac{x_i^T x_j}{\|x_i\|_2 \cdot \|x_j\|_2}$$



- Let's see an example

$$d(x_1, x_2) = 1 - \frac{(1 * 0 + 1 * 2 + 2 * 2 + 1 * 0 + 0 * 2)}{\sqrt{1^2 + 1^2 + 2^2 + 1^2 + 0^2} * \sqrt{0^2 + 2^2 + 2^2 + 0^2 + 2^2}} = 0.3453$$

$$x_1 = \begin{bmatrix} 1 \\ 1 \\ 2 \\ 1 \\ 0 \end{bmatrix} \text{ and } x_2 = \begin{bmatrix} 0 \\ 2 \\ 2 \\ 0 \\ 2 \end{bmatrix}$$

Distance Measurement Types

- Mahalanobis distance

- MD is the distance between two points in multivariate space, represented by d-dimensional feature vectors x_i, x_j

$$d_{Mahalanobis}(x_i, x_j) = \sqrt{(x_i - x_j)M^{-1}(x_i - x_j)^T}$$

where, M is the covariance matrix of the data.

- Intuitively, the covariance matrix generalizes the notion of variance to multiple dimensions
- MD can be thought of **scaling each data dimension by its variance** and adjusting for their relationships
- When **data are independent**, i.e. $M=I$ (identity matrix), Mahalanobis distance becomes same as Euclidean distance.

Distance Measurement Types

- Jaccard distance

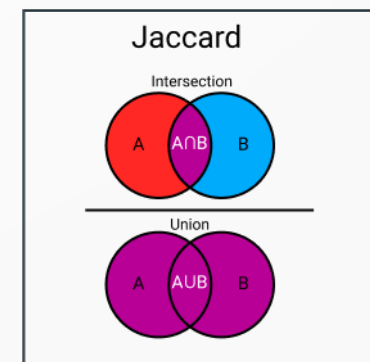
- Is a distance used to measure diversity of **any two sets**
- Consider any two instances x_i and x_j as **binary vectors indicating presence or absence of features**

- Jaccard distance between x_i and x_j is defined as
$$d_{Jaccard}(x_i, x_j) = 1 - \frac{|x_i \cap x_j|_1}{|x_i \cup x_j|_1}$$

- Where $\cap$ denotes logical 'AND' and $\cup$ denotes logical 'OR' operators. The $|x|_1$ is 1-norm

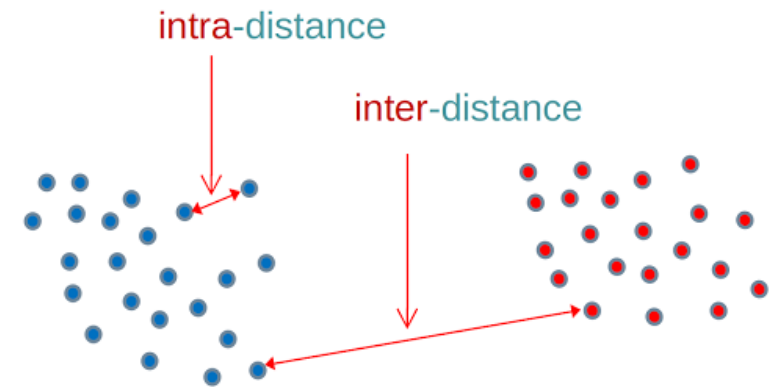
- Jaccard distance for $x_i = [1,0,1]$, and $x_j = [1,1,0]$

$$d_{Jaccard}(x_i, x_j) = 1 - \frac{1}{3} = \frac{2}{3}$$



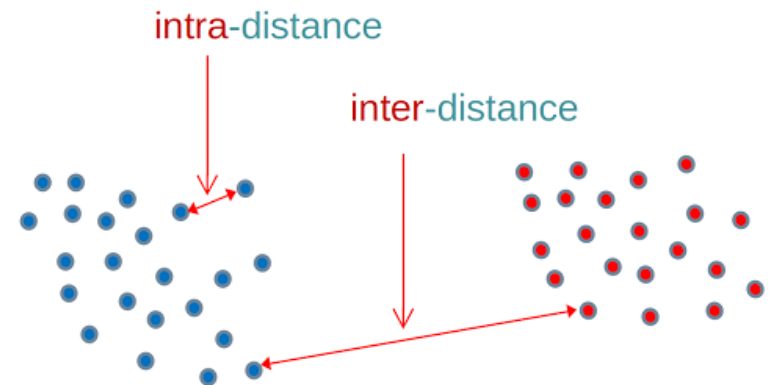
Clustering Algorithms

- How do we teach a computer to do this?
- Goal of clustering algorithms are:
 - Group objects of similar properties together
 - Discover interesting clusters and groups in the data
 - Find valid organisation of the data
- In other words, we can define two algorithmic goals:
 - **Minimise intra-distance** (distance between points in the same cluster)
 - **Maximise inter-distance** (distance between points from different clusters)



Clustering Algorithms...

- Now we can define a generic set-up based on our current understanding from clustering methods:
 - Step 1: define a distance metric between objects
 - Step 2: define an objective function that gets us to our clustering goal
 - Step 3: devise an algorithm to optimise the objective function



How Kmeans Works

- The most popular clustering algorithm; simple and fast
- Kmeans
 - stores k centroids
 - A point is considered to be in a particular cluster if it is closer to that cluster's centroid than any other centroid.
 - KMeans searches for the best centroids by alternating between two methods:
 - Assigning data points to clusters based on the current defined centroids (points which are the centre of a cluster).
 - Choosing centroids based on the current assignment of data points to clusters

N.B. Please see the video presented in “3.5 How Kmeans works” for better understanding

Evaluation of Clustering

- ❖ External assessment:
 - ❖ Compare clustering performance against a known clustering (often called Ground truth or Gold standard)
- ❖ Internal assessment:
 - ❖ Determine if clustering follows certain intrinsic assumptions (e.g. cluster-to-cluster distance or cluster size etc.),

Evaluation of Clustering...

- The following figure illustrates a sample of ground truth (C) and the clustering partition found by a clustering algorithm (C')



Ground truth (C)



Clustering partition found by algorithm (C')

Evaluation of Clustering...

- ❖ Rand Index

- ❖ Is a measure of the similarity between two data clusters
- ❖ Rand index is a function that measures the similarity of the two assignments C and C', ignoring their permutations.

- ❖ Rand index is computed as
$$R = \frac{a + b}{\binom{n}{2}} = \frac{a + b}{a + b + c + d}$$

- ❖ a = the number of pairs of data instances that are in the same cluster in both C,C'
- ❖ b = the number of pairs of data instances that are in the different clusters in C and in different clusters in C'
- ❖ c = the number of pairs of data instances that are in the same cluster in C but in different clusters in C'.
- ❖ d = the number of pairs of data instances that are in the different clusters in C but in the same clusters in C'

Evaluation of Clustering...

- Purity

- Measure the purity for all clusters in terms of class labels of the data in each cluster i.e.,

$$Purity = \frac{\sum_{i=1..q} A_i}{\sum_{i=1..q} n_i}$$

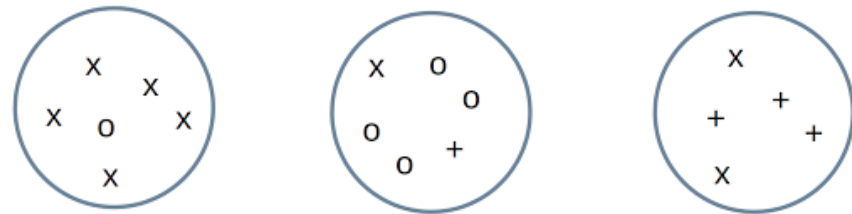
where q is the number of clusters, A_i is the number of correctly assigned elements in i^{th} cluster and n_i is the number of elements in each cluster.

- This means the purity is measured by counting **the number of correctly assigned instances** and **dividing by the number of total instances**.

Evaluation of Clustering...

❖ Purity Example

- ❖ Let's we have three clusters as shown in the figure.
- ❖ Each cluster is assigned to the class label which has the majority in the cluster i.e., first cluster labelled as Cross, second as Circle and last one as Plus.
- ❖ Based on the figure, 5 crosses, 4 circles, 3 pluses were correctly assigned.



$$Purity = \frac{1}{17} * (5 + 4 + 3) \approx 0.71$$

Evaluation of Clustering...

- Mutual Information
 - Mutual information is a function that measures **the agreement of the two clustering assignments** C and C' in terms of how informative one is about the other, ignoring permutations.
 - To put it simple, how informative is C about C'
 - Let us assume that clustering partition C has K clusters and the partition C' has K' clusters

$$MI(C, C') = \sum_{i=1}^K \sum_{j=1}^{K'} P(i, j) \log \frac{P(i, j)}{P(i)P'(j)}$$

where $P(i)$ denotes the probability of randomly selected instance to belong to i^{th} cluster of the partition C, similarly, $P(i, j)$ denotes the probability of a randomly selected instance to belong in i^{th} cluster of the partition C and j^{th} cluster of the partition C'. $P'(j)$ is defined similarly as $P(i)$.

- If our **C' clustering is highly informative based on C**, we can conclude that the C clustering assignment is doing good

Evaluation of Clustering...

- Silhouette Coefficient
 - It is a measure of **how similar an object is to its own cluster** (cohesion) compared to other clusters (separation)
 - This method **does not require the ground truth cluster** assignments
 - The silhouette coefficient contrasts the average distance between the instances of the same cluster with the average distance between the instances of different clusters

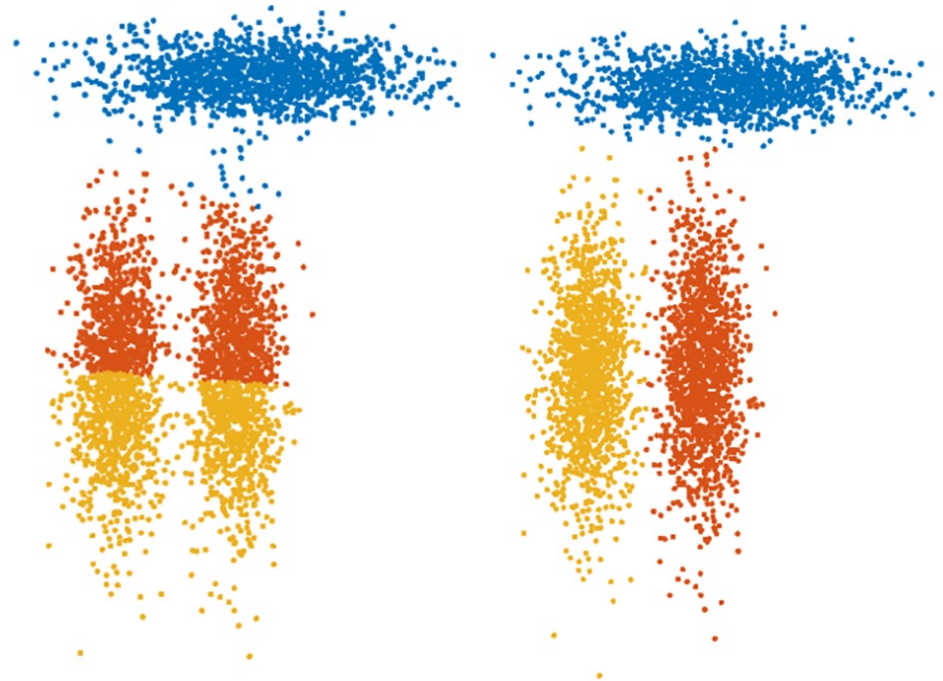
$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}$$

Where, $a(i)$ is the average distance of i^{th} instance with all other instances of the same cluster, and $b(i)$ is the lowest average dissimilarity of i^{th} instance with all other clusters.

- The final value of Silhouette ranges from -1 to $+1$. **High value of $s(i)$ indicates that the object is well matched to its own cluster and poorly matched to neighbouring clusters.**

Limitations of Kmeans

- ❖ Most important limitations of Simple Kmeans are:
 - ❖ Random initialisation means that you may get different clusters each time.
 - ❖ We have to supply the number of clusters beforehand.
 - ❖ It cannot find clusters of arbitrary shapes.
 - ❖ It cannot detect noisy data points.

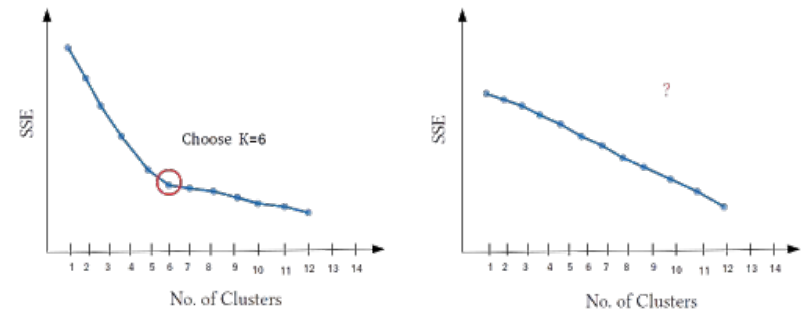


Finding the Number of Clusters

- Elbow Method
 - The idea of elbow method is to run Kmeans clustering algorithm for a range of values of K, and for each value of K, compute the sum of squared error (SSE) as:

$$SSE_k = \sum_k \sum_i z_{ik} \|x_i - \mu_k\|^2$$

- So, $\|x_i - \mu_k\|^2$ finds the distance of each point (x_i) to its corresponding centroid (μ_k) in the cluster
- z_{ik} is a binary variable
 - 1 when x_i is assigned to the cluster number k and
 - 0 when x_i is not related to cluster number k
- it looks 6 is the best cluster number



Clustering with Kmeans++

- Kmeans++ is an algorithm for choosing the initial cluster's centre values or centroids for the Kmeans clustering algorithm
- K-means++ starts with allocating one cluster centre randomly and then searches for other centres given the first one
 - Choose one centroid μ_1 uniformly at random from dataset
 - Let $D(x)$ be the distance from a data point to the closest centroid we have already chosen.
 - Choose a new centroid from the dataset with probability of $\frac{D^2(x_i)}{\sum_i D^2(x_i)}$
 - Now repeat previous step until we have initialised K centroids

Other Clustering Algorithms...

Partition based clustering	K-means clustering CLARANS (Clustering Large Applications based upon Randomized Search)
Hierarchical based clustering	CURE (Clustering Using Representatives) BIRCH (Balanced Iterative Reducing Clustering and using Hierarchies)
Density based clustering	DBSCAN (Density-based Spatial Clustering of Applications with Noise) OPTICS (Ordering Points to Identify Clustering Structure)
Grid based clustering	STING (Statistical Information Grid) Wave cluster
Model based clustering	MCLUST (Model-based Clustering) GMM (Gaussian Mixture Models)

Thank You.

